www.tntextbooks.in

contain carbohydrates and fats for energy yielding, proteins for growth and replacement; and vitamins, minerals and water for physiological regulation.

Vitamins:

Vitamins are naturally occurring organic substances regularly needed in minute quantities for maintaining normal health as metabolic regulators. The identified vitamins are classified as fat soluble (A,D,E and K) and vitamin B and vitamin C are water soluble. Vitamin A, D, E and K, if consumed beyond required level may cause defects, commonly referred to as hypervitaminosis.

Minerals:

These are the inorganic chemical elements, i.e., Ca, Fe, I, K, Mg, Na, P, S, etc needed for regulation of various physiological functions. These can be classified into major minerals (Na, P, K, Ca, Mg, S, Cl) and others are trace minerals such as Fe, Cu, Zn, Co, Mn, I, and fluorine. Sodium ions are more abundant than any other cation in the body fluids.

N.I. Lunin discovered vitamins but the name vitamin was given by Dr. Funk (1912). The first vitamin isolated was B1 by Dr.Funk. The first vitamin produced by fermentation process using, *Acetobacter* bacteria is Vitamin C.

Table 1. Fat soluble vitamins

Vitamins	Functions	Symptoms of Deficiency
A (Retinol)/ Antixerophthalmic vitamin	Plays a vital role in visual perception. Maintenance and growth of epithelial tissue.	Night blindness **(Nyctalopia)**, Xerophthalmia (drying of eyeballs), **Bitot's spot** in the cornea, **Dermatosis** (dry and scaly skin) and **Keratomalacia** Atrophy of lacrymal glands and reduction in tear secretion
D (Calciferol)/ Antirachitic vitamin	Promotes intestinal absorption of calcium and phosphorus. Formation of teeth and bones.	**Rickets** in children (softness and deformities of bones and bow legs and pigeon chest) and **Osteomalacia** in adults (weak and fragile bones, bent, deformed pelvis).
E (Tocopherol) / Antisterility vitamin	Antioxidant It keeps the skin healthy by reduces the process of ageing.	**Sterility** in animals, Ruptured red blood cells
K Anti haemorrahagic vitamin.	1. Required for the synthesis of prothrombin in the liver.	Defect in blood clotting called Haemorrhagic manifestations.